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Review Article

Effectiveness and Safety of Intravenous Medications for the Management of Acute Disturbance (Agitation and Other Escalating Behaviors): A Systematic Review of Prospective Interventional Studies

Megan Castro, M.Sc.,[1] Matt Butler, M.D.,[1] Alysha Naomi Thompson, B.Sc., Siobhan Gee, Ph.D., Sotiris Posporelis, M.D.

Acute disturbance is a broad term referring to escalating behaviors secondary to a change in mental state, such as agitation, aggression, and violence. Available management options include de-escalation techniques and rapid tranquilization, mostly via parenteral formulations of medication. While the intramuscular route has been extensively studied in a range of clinical settings, the same cannot be said for intravenous (IV); this is despite potential benefits, including rapid absorption and complete bioavailability. This systematic review analyzed existing evidence for effectiveness and safety of IV medication for management of acute disturbances. It followed a preregistered protocol (PROSPERO identification CRD42020216456) and is reported following the guidelines set by Preferred Reporting Items for Systematic Review and Meta-Analysis. APA PsycINFO, MEDLINE, and EMBASE databases were searched for eligible interventional studies up until May 30th, 2023. Data analysis was limited to narrative synthesis since primary outcome measures varied significantly. Results showed mixed but positive results for the effectiveness of IV dexmedetomidine, lorazepam, droperidol, and olanzapine. Evidence was more limited for IV haloperidol, ketamine, midazolam, chlorpromazine, and valproate. There was no eligible data on the use of IV clonazepam, clonidine, diazepam, diphenhydramine, propranolol, ziprasidone, fluphenazine, carbamazepine, or promethazine. Most studies reported favorable adverse event profiles, though they are unlikely to have been sufficiently powered to pick up rare serious events. In most cases, evidence was of low or mixed quality, accentuating the need for further standardized, large-scale, multi-arm randomized controlled trials with homogeneous outcome measures. Overall, this review suggests that IV medications may offer an effective alternative parenteral route of administration in acute disturbance, particularly in general hospital settings.

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Key words: acute disturbance, intravenous, midazolam, diazepam, lorazepam, olanzapine, haloperidol, droperidol, ketamine, promethazine, dexmedetomidine, clonidine, clonazepam, valproate, systematic review.

INTRODUCTION

Acute disturbance is a broad term referring to escalating behaviors secondary to a change in mental state. These behaviors include restlessness, agitation, aggression, violence, verbal or nonverbal threats, and intent to cause harm or damage. Underlying factors may include

[1]Joint first authors.

Received August 16, 2023; revised January 15, 2024; accepted January 28, 2024. From the Neuropsychiatry Research and Education Group, Institute of Psychiatry (M.C., M.B., S.P.), Psychology, and Neuroscience, London, United Kingdom; South London and Maudsley NHS Foundation Trust (M.B., S.P.), London, United Kingdom; GKT School of Medical Education, King's College London (A.N.T.), London, United Kingdom; Pharmacy Department (S.G.), South London and Maudsley NHS Foundation Trust, London, United Kingdom; and Institute of Pharmaceutical Sciences (S.G.), Faculty of Life Sciences and Medicine, KCL, London, United Kingdom. Send correspondence and reprint requests to Matt Butler, MD, King's College London Institute of Psychiatry Psychology & Neuroscience, London, United Kingdom; e-mail: matthew.butler@kcl.ac.uk

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psychiatric or physical disorders such as dementia, psychosis, stroke, and substance intoxication.[1]

Acute disturbance is common. A 2015 survey indicated that up to 80% of emergency department (ED) staff experienced workplace violence in the preceding year, with the majority of these incidents perpetrated by patients.[2] A meta-analysis found up to 1 in 5 psychiatric inpatients may commit a violent act by causing, or attempting to cause, harm to another person.[3] The exact prevalence is hard to quantify, in part due to heterogenous definitions. The impact of acute disturbance is serious. Violence and aggression in healthcare settings risk physical and psychological harm to patients, staff, and families.[4] While emergency medicine and psychiatric wards are most commonly impacted, evidence suggests its effect is widespread across healthcare settings.[5]

Despite being common and serious, there is a lack of consensus on the management of acute disturbance.[6] Nonpharmacological practices are agreed upon first-line interventions,[7] with the objective of achieving a state of calmness via verbal engagement, establishment of a collaborative relationship, and de-escalation.[8] Nevertheless, in certain situations, sedation may be unavoidable or preferable, particularly when the safety of the patient and/or others is at risk.[1] There are several established and well-researched pharmacological options for management of acute disturbance, though recommendations vary between guidelines.

Medications may be administered via various routes including oral, intramuscular (IM), buccal, or intravenous (IV). Each route has its own advantages and disadvantages. Choosing the most appropriate can be a complex decision depending on a variety of clinical, pharmacological, and environmental factors.[1,7] While oral and IM are more widely used, there is a clear role for IV, as outlined in recommendations posited by the Royal College of Emergency Medicine as well as the British Association for Psychopharmacology.[1,9,10] IV medications may be administered as a bolus (a single dose over a short time) or as a steady infusion over shorter or longer periods. Supplementary Table 1 summarizes these guideline positions on parenteral medication for acute disturbance.

In comparison to oral and IM, the IV route has some potentially significant advantages. It produces a more rapid effect and leads to complete bioavailability, meaning smaller doses are required to produce desired effects. It also allows for greater control over dosing (e.g., it can be administered at a constant rate), precise titration, and bypass of hepatic metabolism.[11] Conversely, IV medication has shortfalls; rapid absorption and high bioavailability of the medication may lead to a greater risk of significant side effects, and establishing IV access in a highly agitated patient is often challenging.[12] Despite the increasing interest, the research data on the safety and effectiveness of IV medications for acute disturbance has yet to be systematically reviewed.

The aim of this systematic review was to identify and evaluate published evidence on the safety and efficacy of IV medication in the management of acute disturbance via prospective interventional studies.

METHODS

Search Strategy

We followed the definition of acute disturbance in ref[1] as "a composite term [which] includes the concepts of 'agitation', 'aggression', and 'violence'". Specifically, we looked to summarize data on the IV administration of predefined commonly used medications in adults with acute disturbance (agitation, aggression, or violence) in hospital or prehospital settings. Outcomes of interest were any relevant pre- and post-safety and effectiveness measures relating to sedation.

A search of MEDLINE, EMBASE, APA, and PsycINFO was conducted until May 30th, 2023 (no lower limit) via OvidSP. Medical subject heading (MeSH) and free-text search terms were agreed upon with a university librarian. Search terms used: (agitation OR aggression OR violence) AND (midazolam OR diazepam OR lorazepam OR olanzapine OR haloperidol OR droperidol OR ketamine OR promethazine OR dexmedetomidine OR clonidine OR clonazepam OR valproate) AND (IV). Following peer review, a search was re-run to include (diphenhydramine OR propranolol OR ziprasidone OR fluphenazine OR chlorpromazine OR carbamazepine). Additionally, reference lists were searched for relevant publications. Abstracts and full texts were reviewed by M.C. and A.T./M.B. Any disagreements were settled by S.P. and/or M.B., and a consensus was reached. A detailed description of the methods can be found in Supplementary Information. This review was registered on PROSPERO (identification CRD42020216456) and is reported via the Preferred Reporting Items for Systematic Review and Meta-Analysis.

Inclusion and Exclusion Criteria

Articles were included if they described prospective interventional studies of medications administered via the IV route explicitly for acute disturbance in adults. We included papers with delirium; however, this was only when the specific indication for the administration of IV medications was for acute disturbance, and outcomes include sedation-related measures. Only full peer-reviewed articles were included.

Articles were excluded for the following reasons: noninterventional or retrospective designs, participants <18 years, non-English language, other routes of administration (i.e., IM, subcutaneous, oral), premedication for agitation, medication in the context of mechanical ventilation, and any other indication aside from acute disturbance. Studies were excluded if the indication for prescribing medications was vague or not specified for acute disturbance specifically; this included any medical or dental procedures, mechanical ventilation (including weaning), intubation, epilepsy, and pain. Studies that involved premedicating for agitation, such as medication administered before an operation to pretreat emergence agitation or postoperative delirium, were also excluded, as this was considered as a prophylactic measure (i.e., to reduce the occurrence risk of acute disturbance) rather than management of acute disturbance per se.

Quality Assessment

Quality assessment was performed by MC and MB. The National Institutes of Health checklist for quality assessment of controlled interventional studies was used to assess quality.[13] The GRADE criteria were used to measure confidence in the evidence.[14]

Data Extraction

Data from each study extracted by MC and MB included (where applicable): study design, clinical setting, number of participants, indication, medication, dosage, and safety and effectiveness outcome measures.

RESULTS

Search Results

Initial search produced 5603 results. Filter implementation and deduplication removed 1692 articles. Titles and abstracts (3,911) were reviewed, leading to 105 full-text screens. Fifteen studies eventually met the criteria for inclusion (Figure 1). We did not identify any eligible articles on the use of IV clonazepam, clonidine, diazepam, promethazine, diphenhydramine, propranolol, ziprasidone, fluphenazine, or carbamazepine. Given the methodological heterogeneity of included studies, we were unable to pool outcomes for analysis; hence, results are reported as a narrative synthesis.

Outcome measures are summarized in Table 1. Scales used to measure sedation, agitation, or combativeness are presented in Supplementary Table 2.

Quality Assessment

On the National Institutes of Health quality checklist, seven studies were rated as good quality, two as fair, and six as poor (Supplementary Table 3).

On the GRADE criteria checklist, the certainty of evidence from one study was rated as high, four were rated moderate, and ten were rated very low (Supplementary Table 4).

Settings

Ten studies took place in ED settings, two in intensive care units, two in palliative care settings, and one each in a pre-hospital setting and psychiatric inpatient unit. All medications were prescribed specifically for the treatment of acute disturbance.

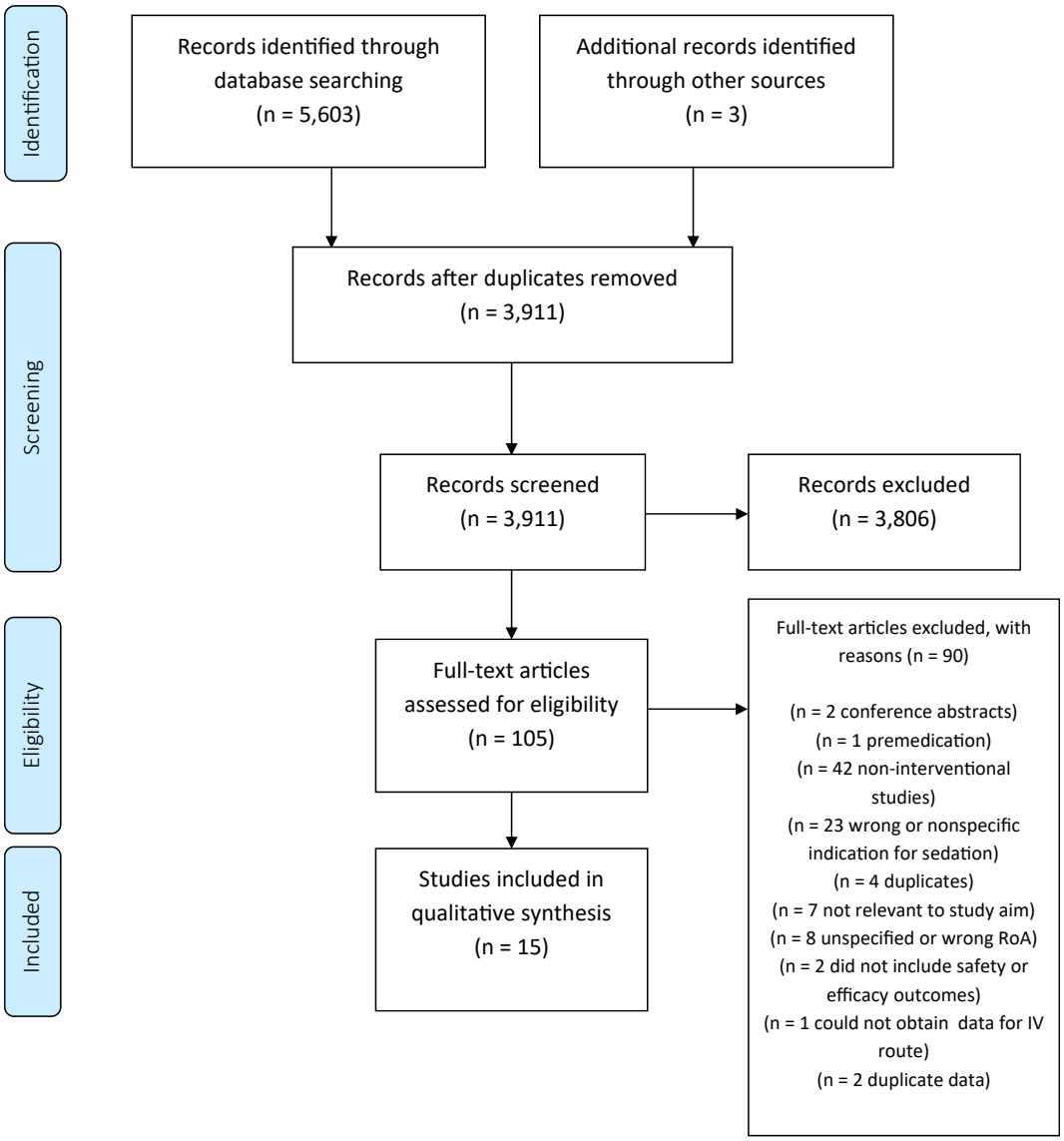
Effectiveness and Safety

Antipsychotic Medications

Droperidol. We found seven studies that examined droperidol. Droperidol significantly outperformed placebo in terms of sedation response[15] and requirement for additional sedative agents.[16] Droperidol alone had an average sedation time of eight minutes.[17] In two studies, droperidol was combined with midazolam; time to sedation in this combination was an average of 5–6 minutes.[18,19] Droperidol outperformed lorazepam for sedation levels at 15–60 minutes[20] and was equivalent to haloperidol at 5–60 minutes.[21] In all seven studies, no significant side effects occurred. Minor side effects (see Table 1) were easily managed by simple clinical interventions.[15–21]

Olanzapine. We found two studies examining olanzapine, both comparing it to droperidol. In one study, patients who received olanzapine had a significantly

FIGURE 1. PRISMA Flow Diagram for Included Studies. Note. This diagram demonstrates the details and reasoning behind the inclusion process for studies in the systematic review. The included numbers are the sum of both searches (please see Methods section for further information). RoA = route of administration; PRISMA = Preferred Reporting Items for Systematic Review and Meta-Analysis



shorter mean time to adequate sedation compared to droperidol and control groups; in all other measures, olanzapine and droperidol were equivalent.[18,19] QTc prolongation was not a concern, and olanzapine was overall deemed to be safe with only very few adverse events.[18,19]

Haloperidol. We found four studies examining haloperidol. Haloperidol produced significant improvements in sedation scores.[22,23] Time to sedation ranged from 10–115 minutes in three patients who received a combination of IV haloperidol-lorazepam.[24] In one study, a haloperidol infusion following pretreatment with boluses of haloperidol was inferior to a dexmedetomidine infusion.[25] In one palliative care study, haloperidol was equivalent to chlorpromazine for sedation scores. Some adverse events, including death, were reported; however, these were not unexpected in the patient population.[22] In one other study, a patient in the haloperidol group had QTc >500 ms. No other serious adverse events occurred.[23]

Chlorpromazine. We found one study examining chlorpromazine (see above).[22]

TABLE 1. Interventional Studies for Efficacy of Intravenous Medications Used for Acute Disturbance

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Carrasco et al. (2016)[25]	Quasi-Experimental	ICU (nonintubated)	Very low	132	DEX: 70.3 HAL: 71.3	DEX: 80% HAL: 89%	Unknown	DEX 0.2 μg/kg/h HAL 2.5–5 mg (bolus); 0.5–1 mg/hour (maintenance)	DEX: B, SI HAL: B, SI All received IV boluses of HAL until a) effective sedation OR b) max 30 mg dose. Responders a) were then given a HAL infusion, whereas the nonresponders b) were given DEX boluses until adequate sedation, followed by an infusion of DEX.	Quality of Sedation - % of time patient was maintained at adequate level of sedation using RASS scores	Median % of time patients spent under adequate sedation – DEX: 92.7% (95% CI 84.5–99.8%); HAL: 59.3% (95% CI 49.1–78.0%) (P = 0.0001)	Median Percentage of Patients with AEs: Excessive Sedation (DEX – 0%; HAL – 11.6% (95% CI 6.5–21.2%) (P = 0.01) None in the DEX group experienced excess sedation or treatment failure. Five had bradycardia (responsive to atropine), and 12 had a transient supraventricular arrhythmia. These rates were not different to the IV haloperidol group.
Calver & Isbister (2012)[28]	Single-arm, pre-post study	ED	Very low	13	DEX: 44	DEX: 85%	Unknown	DEX 1 mg/kg (loading dose); 0.7 μg/kg/h (infusion)	DEX: B, SI All were given a loading dose of DEX, then stratified by a) effective sedation, b) repeated loading dose required (agitation reoccurred after being effectively sedated) or c) infusion of DEX required as not sedated	Adequate sedation (defined as Sedation Assessment Tool score reduction from +2 or+3 by two levels or returning to zero for a period of >1h) Frequency of Adverse Events – need for airway support	5/13 patients were adequately sedated with DEX; 2 with loading dose, 3 with loading dose + infusion.	6/13 patients developed hypotension, in which majority occurred during the infusion of DEX rather than when loading doses were given. Respiratory compromise did not occur in patients.

TABLE 1. (*Continued*)

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Rosen et al. (1997)[15]	Prospective, Randomized, double-blind	Pre-hospital	Very low	46	DROP: 34 SALINE: 36	DROP: 100% SALINE: 100%	Unknown	DROP 5 mg	DROP: UNK All patients received the contents of a 2-cc vial on scene	Five-point agitation scale score	Agitation scores 5 mins postadministration: DROP (2.09); CTRL (2.78) ($P = 0.05$) −10 minutes postadministration: DROP (1.04); CTRL (2.57) ($P < 0.001$)	One instance of akathisia was reported in the droperidol group which responded to antihistamine medication. No serious adverse events, neuroleptic malignancy syndrome, or hypotension occurred.
Thomas et al. (1992)[21]	Randomized double-blind RCT	ED	Very low	68	DROP: 36 HAL: 31	DROP: 83% HAL: 100%	Unknown	DROP 5 mg HAL 5 mg	DROP: UNK HAL: UNK All were given DROP or HAL until either a) effective sedation was achieved or b) a repeat dose or other psychotropic was given (if not effectively sedated after 30 mins)	Combativeness scale, scored up to 60 mins postadministration.	Combativeness Score 10 minutes postadministration: HAL (2.24); DROP (3.26) −60 minutes postadministration: HAL (4.39); DROP (4.42) ($P = 0.78$) (Twelve patients received IV haloperidol, and nine received IV droperidol (the remaining 47 received IM haloperidol/ droperidol and were not included in this review). Lack of baseline scores renders this study difficult to fully interpret.	No serious adverse events occurred in IV patients.

TABLE 1. (*Continued*)

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Taylor et al. (2017)[18]	Three arm blinded RCT	ED	High	349	DROP: 34 OLA: 35 MZ-DROP: 34	DROP: 68% OLA: 69% MZ-DROP: 72%	Unknown	DROP 10 mg OLA 10 mg MZ 5mg + DROP 5 mg	DROP: B OLA: B MZ-DROP: B All were given boluses of respective IV meds until either a) effective sedation was reached or b) they received up to 2 additional 5mg boluses	Proportion of patients sedated adequately (using six-point sedation scale) at10 mins postadministration (adequate sedation = score of 2 or less)	% of Patients Maintained at Sedation Level – 10 minutes postadministration: MZ-DROP (88 pts; 74.6%); DROP (55 pts; 409.6%); OLA – 59 pts (49.2%) - The difference in proportions from MZ-DROP was 25% compared to DROP (95% CI 12.0% to 38.1%) and 25.4% to OLA (95% CI 12/7% to 38.3%). Median time to satisfactory sedation: MZ-DROP 5 mins, DROP 11 mins, OLA 11 mins.	Proportion of Patients with AEs – MZ-DROP (26 pts; 22.0%); DROP (18; 16.2%); OLA (24; 20.0%) - All adverse events were readily managed in the clinical setting (respiratory depression was readily controlled without intubation). - In the DROP only group the following AEs were reported: SaO2 <90% 6.3%, systolic blood pressure <80 in 3.6%, heart rate <60bpm n 1.8%, QTc >500ms in 2.7%, and respiratory rate <10 breaths/min in 0.9%. No instances of acute dystonia were reported.
van Leeuwen et al. (1977)	Quasi-RCT	Psych.	Very low	19	Median age (39.5 (DROP) & 27.5 (PLACEBO))	DROP: 84% PLACEBO: 47%	Unknown	DROP 10 mg	DROP: UNK PLACEBO: UNK All received an IV dose, then another dose of 5mg HAL if not controlled within 3 mins.	Rescue treatment with HAL	3 minutes post DROP/Placebo injection – 6/19 patients in the DROP group needed HAL rescue - 19/22 patients in the CTRL group needed HAL rescue. After 30 mins, four patients in the droperidol group and 10 in the placebo group needed further individually tailored psychotropic medication.	No adverse events occurred.

TABLE 1. (*Continued*)

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Khalil et al. (2019)[23]	NRS	ICU	Very low	100	63 (does not differentiate between groups)	100% (does not differentiate between groups)	Unknown	HAL (unknown dose)	HAL: SI All patients received an IV infusion for 72 hours	RASS Score	RASS score: – HAL initial (2.6 ± 0.67); after (0.34 ± 0.48) ($P < 0.001$) – CTRL initial (2.52 ± 0.61); after (0.28 ± 0.54) ($P < 0.001$)	One patient in the HAL group had QTc prolongation but did not have significant cardiac adverse events. No other significant adverse events occurred.
Hui et al. (2020)[22]	Parallel-group Randomized Trial	Palliative Care	Very low	68	HAL: 65 CHLOR: 63 HAL + CHLOR: 60	HAL: 67% CHLOR: 63% HAL + CHLOR: 43%	HAL – White: 67% Black: 13% Hispanic: 7% Other: 13% CHLOR – White: 81% Black: 12% Hispanic: 6% Other: 0% HAL + CHLOR -White: 71% Black: 14% Hispanic: 14% Other: 0%	HAL 2 mg CHLOR 25 mg HAL 1mg + CHLOR 12.5mg	HAL: SI CHLOR: SI HAL + CHLOR: SI All patients were started on a HAL-only regimen 2 mg every 6 hours, and 2 mg every 1 hour has needed. If patients needed rescue medication, then they were given infusion of HAL, CHLOR, or HAL + CHLOR	Change in RASS Score from 0 to 24 hours	Mean change in RASS score 30 mins postadministration – HAL escalation (−2.6; 95% CI -3.6 to −1.3); CHLOR (−2.4; 95% CI -3.0 to 1.8): HAL + CHLOR (−2.1; 95% CI -3.0 to −1.3) ($P = 0.86$)	Deaths (not otherwise specified)* occurred in 93% of the HAL group, 100% of the CHLOR group, and 100% of HAL + CHLOR There were no reports of akathisia, apnea, or sinus tachycardia.
Lin et al. (2020)[26]	Prospective Randomized Open-label	ED	Very low	93	Not specified for IV	Not specified for IV	Not specified for IV	KET 1 mg/kg HAL + LZ (10 mg HAL + 2 mg LZ)	KET: UNK HAL + LZ: UNK IV patients were administered either IV KET or IV HAL + LZ	% of patients adequately sedated within 5 minutes using the RASS score	% of patients adequately sedated – 5 minutes postadministration: HAL + LZ (0%); KET (0%) Average time to sedation – KET (25.5 mins); HAL + LZ (53.3 mins)	Not specified for IV patients specifically.
Hui et al. (2017)	RCT	Palliative Care	Moderate	90	LZ + HAL: 66 HAL + PLAC: 64	LZ + HAL: 62.1% HAL + PLAC: 44.8%	LZ + HAL – White: 72.4% Black: 13.8% Hispanic: 6.9% Other: 6.9% HAL + PLAC – White: 79.3% Black: 13.8% Hispanic: 0% Other: 6.9%	LZ + HAL 3 mg HAL + PLAC 3 mg	LZ + HAL: SI, BI HAL + PLAC: SI, BI All patients received 2mg IV HAL every 4 hours and additional 2mg every hour if needed; if not controlled, then patients received PLAC or LZ, then HAL	RASS Score	30 mins postadministration, there was a −3.6 reduction in RASS score in the lorazepam group compared with −1.6 reduction in RASS score in the placebo group ($P < 0.001$). RASS Score after 8 hours – LZ + HAL (−4.1 points); HAL + placebo (−1.9 points) ($P < 0.001$)	No significant adverse events occurred. Percentage of patients who developed AEs – Hypokinesia (LZ + HAL: 19%); (HAL: 27%) – Akathisia (LZ + HAL: 19%); (HAL: 7%)

TABLE 1. (*Continued*)

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Richards et al. (1998)[20]	Prospective, Randomized	ED	Very low	202	LZ: 34.6 DROP: 33.2	LZ: 63% DROP: 61%	LZ -Caucasian: 70% DROP -Caucasian: 69%	LZ 2–4 mg DROP 2.5–5 mg	LZ: UNK DROP: UNK All patients were given either IV LZ or IV DROP until either a) effectively sedated or b) given repeat dose at 30 mins	Six-point sedation scale score (mean/SD)	Mean sedation scores 5 minutes postadministration – LZ (4.7 ± 0.6), DROP (4.8 ± 0.7) $P = 0.08$; 10 minutes (4.1 vs 2.8), 15 minutes (3.5 vs 2.0), 30 minutes (2.9 vs 1.6) and 60 minutes (2.5 vs 1.5) [all $P < 0.001$]	One patient in the droperidol group had an acute dystonic reaction 10 min which responded well to IV diphenhydramine. No other significant adverse events occurred.
Knott et al. (2006)[17]	RCT	ED	Moderate	153	Median age MZ: 35 DROP: 32	MZ: 64% DROP: 65%	Unknown	MZ 2.5–5 mg bolus DROP 2.5–5 mg bolus	MZ: B DROP: B All patients were given IV boluses every 5 minutes until effectively sedated	Time to adequate sedation (defined as a score of 2 or less on their agitation scale)	Median time to sedation – MZ = 6.5 mins (5 mg median dose); DROP = 8 mins (10 mg median dose) $P = 0.075$ % of adequately sedated patients after 5 minutes – MZ (44.6%); DROP (16.5%) ($P < 0.001$)	% of Patients with AEs – Airway Management: MZ (4.1%); DROP (0%) Arrhythmia: MZ (0%); DROP (1.3%) Hypoxia: MZ (5.4%); DROP (3.8%) DROP: 3.8% hypotension, 3.8% dystonia. MID: 5.4% hypotension.
Spain et al. (2008)[27]	Prospective pilot study, convenience sampling	ED	Very low	62	Not specified for IV only	Not specified for IV only	Unknown	MZ 10 mg	MZ: UNK All patients were given either IV or IM MZ every 10 minutes, up to four doses until a) effectively sedated or b) failed safety criteria or c) required rescue medication	% of patients adequately sedated at in 10-minute intervals	% of patients adequately after 10 minutes postadministration: IV MZ (57%); IM MZ (73%) 20 minutes postadministration: IV MZ: 5/8 patients (63%); IM MZ: 2/3 patients (67%). Rescue medication (defined as further administration of midazolam after 60 minutes of the last effective dose) was more frequently used in the IV group (14%) than in the IM alone group (7%).	Incidence of failed safety criteria (airway obstruction, slowed respiration rate, decreased GCS levels, Sa02 levels ≤90%, and hypotension) 14.3% after first dose; 25.0% after the second dose.

TABLE 1. (*Continued*)

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Chan et al. (2013)[19]	Placebo-controlled double-blind RCT	ED	Moderate	336	Median age MZ: 35 DROP + MZ: 32 OLA + MZ: 34	OLA + MZ: 63.3% DROP + MZ: 58% MZ: 55.7%	Unknown	OLA + MZ 5 mg DROP + MZ 5 mg MZ 2.5–5 mg	OLA + MZ: B DROP + MZ: B MZ: B All patients received adjunct bolus of either OLA, DROP, or placebo respective to group assignment then incremental MZ until effective sedation achieved.	Time to adequate sedation (mean & median)	Mean Time to Sedation – OLA + MZ (14.0 min, 33.3 SD); MZ (67.8 min, 197.5 SD); DROP + MZ (21.3 min, 97.1 SD) – Median Time to Sedation –OLA + MZ (5 min, IQR 3–10) DROP + MZ (6 min, IQR 3–10); MZ (10 min, IQR 4–25) At 10 minutes, 67.9% in the olanzapine group were adequately sedated in comparison to 66.1% in the droperidol group and 48.7% in the control group.	- Proportion of patients with AEs – OLA + MZ (8.3%); DROP + MZ (10.7%); MZ (15.7%) – Minor adverse events were seen in this study, and no significant adverse events occurred. - DROP: 2.7% airway obstruction, 8.0% oxygen desaturation, 3.6% hypotension. - OLA: 2.8% airway obstruction, 4.6% oxygen desaturation, 2.8% hypotension, and 0.9% arrhythmia (n = 1, a wide complex bradycardia and left bundle branch block in a patient who had consumed a significant amount of alcohol and heroin). - The median length of QTc intervals was similar in the olanzapine (448 msec), droperidol (441 msec), and control (444 msec) groups

TABLE 1. (*Continued*)

Author & year	Study design	Clinical setting	Quality of evidence (GRADE)	# Of pts	Mean age (years)	Males (%)	Ethnicity	IV Meds + dosage (mean)	Type of administration	Primary outcome measures	Effectiveness results	Safety outcome measures
Asadollahi *et al.* (2015)[29]	RCT	ED	Moderate	80	VAL: 42.95	VAL: 66%	Unknown	Valproate (20 mg/kg)	VAL: SI All patients received continuous IV infusion for 10 minutes until a) effective sedation achieved or b) repeat dose administered	Agitation at 30 minutes postadministration using the Agitation–Calmness Evaluation Scale (ACES).	Mean postintervention ACES score was significantly more (mean 4.73) than baseline (mean 1.64).	In total, 24/80 (30.0%) experienced at least one adverse event, including vomiting (13/80, 16.2%), headache (9/80, 11.2%). No patients experienced extra-pyramidal side effects or hypotension.

B = bolus; BI = brief infusion; CHLOR = chlorpromazine; CTRL = control group; DEX = dexmedetomidine; DROP = droperidol; DZ = diazepam; ED = emergency department; HAL = haloperidol; ICU = intensive care unit; IV = intravenous; IM = intramuscular; KET = ketamine; LZ = lorazepam; Meds = medications; MZ = midazolam; NRS = nonrandomized study; OLA = olanzapine; Pall Care = palliative care; Pts = participants; Psych. = psychiatry; RCT = randomized clinical trial; RASS = Richmond Agitation-Sedation Scale; SI = steady infusion; SD = standard deviation; UNK = unknown.

A further study by Riddell et al. (2017) analyzed the effects of IV and IM ketamine, which resulted in efficacious outcomes for ketamine-sedating acutely disturbed patients in the ED, but authors did not report separately on each route; we approached the authors, but they were unable to provide the split data, and therefore the study was not included in this review [38].

* As this study took place in a palliative care unit, primary endpoints also included death of the patients. The deaths may not be related to/caused by the medication administration.

Benzodiazepines

Lorazepam

We found three studies examining lorazepam. Lorazepam outperformed placebo as an adjunct to haloperidol in a palliative care setting[26]; however, it was inferior to droperidol.[20] The other study was a combination of IV haloperidol-lorazepam.[24] In the palliative care study, hypokinesia was reported[26]; however, no adverse events were reported in the other lorazepam-alone study.[20]

Midazolam

We found two studies examining midazolam. Median time to sedation was 6.5 minutes.[17] After 10 minutes, 55–57% of patients given midazolam were adequately sedated.[17,27] Both studies reported significant adverse events, such as the requirement for airway management.[17,27]

Dexmedetomidine

We found two studies examining dexmedetomidine, both in populations that had failed to respond to an alternative form of sedation. In one study, there were high rates of adequate sedation, which were lower in a smaller study.[28] Treatable cardiac adverse events were noted; however, given that patients had been pretreated, it is impossible to disambiguate the potentially causative agent.[25]

Ketamine

We found one study examining ketamine. Time to sedation in the two patients who received IV ketamine was 8–43 minutes. As the authors also did not differentiate between routes of administration for adverse events, no conclusions can be drawn on safety from this study.[24]

Valproate

We found one study examining valproate. Valproate was effective in sedating patients in this study; however, one in three experienced at least one adverse event.[29]

DISCUSSION

This systematic review summarizes the evidence on the effectiveness and safety of IV medications to treat acute disturbance. Results showed mixed but positive results for the effectiveness of IV dexmedetomidine, lorazepam, droperidol, and olanzapine. Evidence was more limited for IV haloperidol, ketamine, midazolam, and valproate. Most studies reported favorable adverse event profiles, though they are unlikely to have been sufficiently powered to pick up rare serious events.

One of the key advantages of IV medication, also highlighted in included studies, is their rapid onset; this makes them an ideal choice when acute disturbance is unmanageable and risk to person becomes an imminent concern. Anecdotal evidence from our clinical experience in liaison neuropsychiatry specialist settings suggests a role for IV medication to aid administration of other treatments, as in the case of anti-NMDAR encephalitis; this has also been highlighted in relevant case reports.[30] Prescribing clinicians need to take into account not only the practicalities of administering medication via the IV route (i.e., a requirement for adequately trained staff), but also the availability of monitoring and resuscitation equipment.[1] As such, the use of IV medication in many psychiatric units remains problematic and, in some settings, unfeasible. Nevertheless, they remain a suitable option in many cases, and clinicians may need to consider them particularly in general hospital settings.

The literature on IV medications is sparse, partly because some of the studied medications are used off-label. For others, their indication has been assumed indirectly based on existing evidence documenting safety and effectiveness when administered orally or intramuscularly. There is also a lack of comparative effectiveness studies that could highlight the added benefits (or lack thereof) of IV compared to oral and IM routes. Despite improvements in research methodology, several studies lack clear, validated outcome measures.[31]

This review highlighted rapid and favorable sedative outcomes in those administered IV antipsychotics. Despite previous concerns over the propensity for droperidol to prolong the QTc, this has been consistently rebutted.[18,19,32,33] Studies included in this review emphasize the overall safety of droperidol. This is supported by a recent Cochrane review which concluded that droperidol is both safe and effective.[34] Although droperidol is approved for the prevention of postoperative nausea and vomiting by the Food and Drug Administration in the USA, it is still being used off-label for the management of acute disturbance, which may be an additional barrier to its implementation.[33]

The safety results regarding olanzapine are supported by a recent large retrospective review (n = 713) of patients who received IV olanzapine in the ED, which concluded that severe adverse events were rare and that there was no evidence of QTc prolongation.[35] Historically, IV haloperidol has been linked to prolonged QTc as well as extrapyramidal side effects and neuroleptic malignant syndrome. All of these concerns were reassuringly addressed in a recent systematic review, which emphasized its safety.[36]

The effectiveness data for lorazepam and midazolam in our study suggests favorable outcomes. Some concerns have been raised about the impact of benzodiazepines on respiration, particularly in vulnerable population (e.g., pre-existing respiratory disease, co-administered with opioids or alcohol). Nevertheless, when used as single agents in those without pre-existing vulnerability, many benzodiazepines may have a limited effect on respiratory function. In our review, significant respiratory depression did not occur in participants given lorazepam,[20,26] a finding that corresponds to a recent review.[37] Nevertheless, midazolam appeared to be the medication with the highest rates of respiratory complications.[17,27]

This review suggests that dexmedetomidine could be an effective option for difficult-to-sedate patients in the ED.[25,28] There have been concerns that IV dexmedetomidine may be associated with serious cardiovascular events; however, the evidence has been inconsistent.[38] While the cost of dexmedetomidine is higher than that of some other medications, economic evaluations based on length of stay have been favorable.[25,39] Further large-scale interventional trials are required to assess the safety and effectiveness of IV dexmedetomidine.

Ketamine is commonly chosen as a safer choice of sedative medication because of its limited effects on respiratory drive and laryngopharyngeal reflexes. Two meta-analyses have reported favorably on the efficiency of IM ketamine in the treatment of agitation,[40,41] and it is recommended for acute disturbance by the Royal College of Emergency Medicine.[10] Nevertheless, the data included in this review was limited to a single study.[24] Further large-scale interventional research via the IV route is required.

The use of valproate in the management of agitation in dementia was recently contested by a Cochrane review.[42] A further concern is around its well-replicated links with teratogenicity.[43] Alternative options for the management of acute disturbance appear more suitable.

Future Directions

This review highlighted the necessity for further research. We suggest that future studies assess the effectiveness of IV medications for specific underlying causes of acute disturbance, such as acute intoxication with substances (alcohol and drugs), delirium, or psychiatric disorders. This could allow for the development of specific guidelines mirroring those of other routes of administration.[44] Further evidence of sedation efficacy would be welcomed; however, data on downstream outcomes of interest such as hospitalization length, requirement for further chemical or physical restraint, associated costs (e.g., through increased care requirements), and discharge outcomes would significantly help to contextualize the effectiveness of IV medications in a broader sense.

Limitations

Limitations of the Literature

Acute disturbance is not a standardized term, and terminology varies across international settings. There are inherent difficulties in interventional studies for patients presenting with acute disturbance. The quality of eligible studies varied. A variety of primary outcome measures were used, hindering further statistical analysis, and sedation is an imperfect measure of the effectiveness of medications in the management of acute disturbance.

Limitations of this Review

The medications in the search terms were chosen *a priori* and may not be exhaustive. We limited our exclusion to articles in which medications were prescribed specifically for the treatment of acute disturbance; this may have missed some articles in which medications were administered for cognate presentations, such as delirium, but not specifically for acute disturbance (specifically, 2 recent large RCTs on delirium[45,46] were excluded for these reasons). We decided not to include opioid drugs such as fentanyl, morphine, and hydromorphone; while they certainly have sedative (and analgesic) effects, the primary rationale for use in the intensive care unit is sedation

(and analgesia) rather than management of acute disturbance. We did not include any articles without outcomes relevant to sedation. We included only prospective interventional studies; this led to the exclusion of many retrospective studies, which may have bolstered the findings (however, with attendant limitations of this design).

CONCLUSIONS

In this systematic review, we summarized the literature on the effectiveness and safety of IV medication for acute disturbance. In most cases, evidence was of low or mixed quality, accentuating the need for further high-quality research using homogeneous outcome measures and standardized scales. Despite this, our review suggests that several IV medications may be effective options for the treatment of acute disturbance and that safety outcomes generally do not preclude their use in general hospital settings. We particularly suggest that clinicians consider roles for IV droperidol, dexmedetomidine, lorazepam, and olanzapine. Despite this, their use as medications for acute disturbance in psychiatric hospitals may be limited by availability of equipment for administration and monitoring.

We note that the literature suggests that older safety concerns for droperidol may not have borne out in larger datasets and that (aside from midazolam) the respiratory profiles of IV benzodiazepines—when used as single agents—are generally good. The safety profiles of IV midazolam and valproate suggest alternative medications may be preferable, particularly in vulnerable populations. There was limited available data on some anecdotally commonly used medications such as ketamine and haloperidol; however, these medications may remain suitable options via the IV route.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found at 10.1016/j.jaclp.2024.01.004.

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